EpiKG: End-to-End Epistemic Preservation in Clinical Knowledge Graphs for Assertion-Aware Retrieval-Augmented Generation

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Abstract

Clinical NLP pipelines detect assertion status—negation, uncertainty, family history—with $F_1 > 0.90$ [1], yet no existing system propagates these labels through knowledge graph construction into retrieval-augmented generation. We call this the *epistemic propagation gap*: when assertion labels are discarded during graph construction, downstream generation cannot faithfully represent the epistemic state of clinical findings. We present EpiKG, a clinical knowledge graph that closes this gap with a 7-value assertion schema on every edge, a tri-temporal model (valid time, transaction time, and NLP-asserted temporality), and an intent-aware retrieval layer that dispatches question-type-specific graph traversals. On ClinicalBench ($n = 400$, MIMIC-IV, Claude Opus 4.6), intent-aware KG-RAG achieves 77% overall (+28 pp vs. LLM alone at 50%), improving 8 of 9 categories with zero regressions: change detection rises from 10% to 100% (+90 pp), historical reasoning from 6% to 62% (+56 pp), and family history from 3% to 57% (+53 pp). Brute-force long context fares worse than LLM alone (−8 pp), collapsing to 0% on historical and sequence categories. Cross-model validation reveals that KG-RAG benefit is inversely related to parametric medical knowledge: MedGemma 27B gains +14 pp from the same pipeline vs. Opus's +28 pp, consistent with a knowledge-compensator effect. A two-physician blinded adjudication is preregistered. These results establish the category×condition interaction—not the aggregate—as the right evaluation lens for clinical KG-RAG.

1 Introduction

A physician documents "patient denies chest pain, sister had MI at 45, will consider statin if lipids remain elevated"—a single sentence encoding four epistemic states: a negated symptom, a family-history event, a hypothetical action, and an implicit present condition. Clinical NLP can detect these assertions accurately [2, 1], yet when a RAG system ingests the same note, assertion context is flattened: negation is lost, family attribution collapses onto the patient, and downstream reasoning may conclude the patient *has* chest pain and *had* an MI. We call this the *epistemic propagation gap*—the systematic loss of assertion status as clinical information moves from extraction through storage to retrieval.

The gap persists because no system, to our knowledge, propagates assertion labels beyond the extraction stage into knowledge representation or retrieval. OMOP excludes negated conditions from `CONDITION_OCCURRENCE` [3]; FHIR provides `verificationStatus` for conditions only. Graph-augmented RAG systems—GraphRAG [4], GFM-RAG [5], KARE [6], Medical-Graph-RAG [7]—build KGs that discard the metadata distinguishing "patient has diabetes" from "rule out diabetes." Two KG surveys [8, 9] do not identify assertion preservation as a concern. A parallel *temporal integration gap* exists: clinical events span three time dimensions (valid time, transaction time, NLP-asserted temporality), yet existing systems model at most two [10, 11]. Allen's interval algebra [12] has been formalized as annotation ontologies [13] and modal operators [14], but never as first-class KG edge attributes that participate in retrieval.

We present EpiKG, a clinical KG system that closes both gaps by preserving epistemic status end-to-end from NLP extraction through OMOP mapping, fact construction, KG materialization, and graph-augmented retrieval. Our contributions:

1. **Epistemic preservation invariant.** We formalize the requirement that assertion status be recoverable at every downstream stage, give an information-theoretic bound on the loss from assertion-blind pipelines (Section 3.5, Appendix B), and realize the invariant with a seven-value schema extending i2b2 [2], carried from extraction through OMOP mapping, fact construction, KG materialization, and RAG serialization.

2. **Tri-temporal KG with Allen's interval algebra.** Edges carry valid time, transaction time, and NLP-asserted temporality alongside nine temporal relations (seven from Allen's algebra [12] plus CONCURRENT and UNKNOWN) as first-class attributes, enabling temporal reasoning over longitudinal records.

3. **Five-condition ablation with bookend baselines.** We evaluate five retrieval conditions—from C7 (deterministic KG, no LLM; 3.8%) through C4g (intent-aware KG-RAG; 77.0%, +28 pp vs. LLM alone)—on ClinicalBench and SliceBench. Bookend baselines C6 (brute-force long context; 41.8%) and C7 bracket the design space. Intent-aware KG-RAG improves 8 of 9 categories with zero regressions; neither more context nor structured data alone can substitute for LLM + epistemic KG-RAG.

4. **ClinicalBench, SliceBench, and cross-model validation.** Two benchmarks (400 and 144 questions) across 9 epistemic categories target the propagation gap, with hard longitudinal questions (change, cross-admission, temporal) as the primary endpoint. Cross-model validation on MedGemma 27B reveals that KG-RAG benefit is inversely related to parametric medical knowledge; a two-physician blinded adjudication is preregistered (Appendix O).

The central research question is: *under what conditions does structured epistemic context help, hurt, or break even compared to unstructured retrieval or no retrieval at all?* The answer is decisive: intent-aware KG-RAG lifts aggregate accuracy by +28 pp, improving 8 of 9 categories—change detection (+90 pp), historical reasoning (+56 pp), family-history attribution (+53 pp)—with zero regressions. Brute-force long context (−8 pp) performs *worse* than no retrieval at all. The benefit is model-dependent: MedGemma 27B gains +14 pp from the same pipeline vs. Opus's +28 pp, suggesting that KG-RAG benefit is inversely related to parametric medical knowledge. The magnitude and direction of the effect are both category- and model-specific, producing interactions that aggregate accuracy alone obscures (Section 5).

2 Related Work

We organize prior work along four axes and synthesize the gaps that EpiKG addresses (Table 7, Appendix).

Medical RAG systems. Graph-augmented retrieval has emerged as a leading paradigm for medical QA. GraphRAG [4] introduced community-based summarization; GFM-RAG [5] trained an 8M-parameter graph foundation model across 60 KGs (14M+ triples); KARE [6] adapted community retrieval to clinical decision support; DoctorRAG [15] emulates physician reasoning; MedRAG [16] constructs hierarchical diagnostic KGs; Medical-Graph-RAG [7] links documents via triple graphs. Yet no published medical RAG system models assertion status or temporal context; GFM-RAG and KARE build population-level graphs rather than patient-level graphs from clinical text, and a recent survey [17] confirms no system incorporates epistemic metadata into the retrieval graph.

Clinical KG construction. Multi-LLM KG-RAG [18] uses multi-agent LLM prompting with schema-constrained extraction for oncology, modeling uncertainty via entropy but not tracking assertion classes on edges. AutoRD [19] applies GPT-4 to rare disease literature; RECAP-KG [20] handles messy GP notes; FHIR-Ontop-OMOP [21] generates virtual KGs from OMOP data. All share a common limitation: assertion status is not propagated into the final graph.

Assertion detection. NegEx [22] introduced trigger-based negation detection; ConText [23] extended it with temporality and experiencer. The i2b2/VA shared task [2] formalized a six-class taxonomy; Gul et al. [1] fine-tuned LLMs to 0.962 accuracy. All treat assertion detection as a terminal annotation task: labels are not carried into knowledge graphs or retrieval systems.

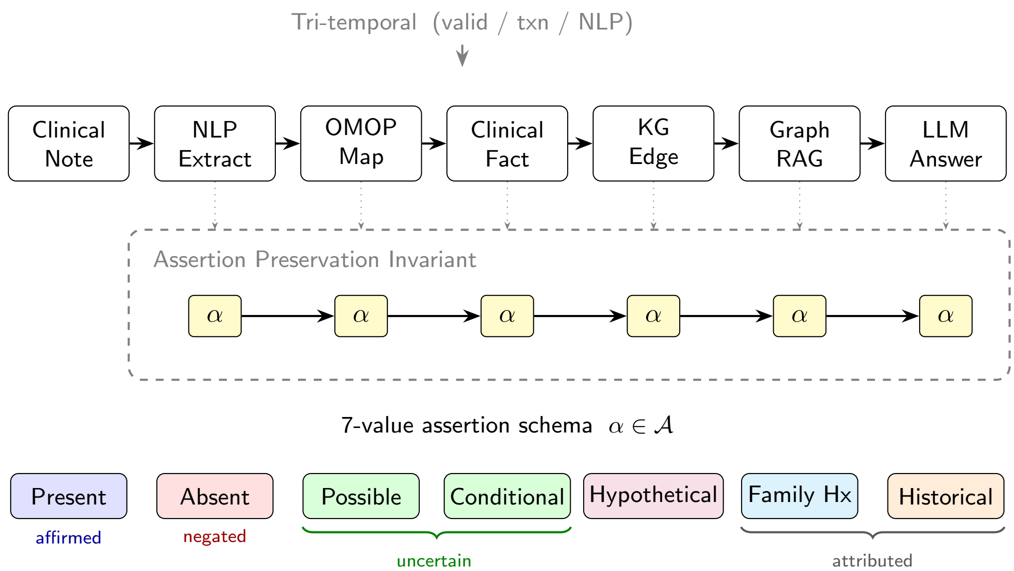

Figure 1: The EpiKG pipeline. The assertion label α (highlighted) is preserved as a first-class attribute from NLP extraction through OMOP mapping, fact construction, KG materialization, and assertion-aware GraphRAG.

Assertion detection is necessary but not sufficient: labels must also be *temporally situated*—the same condition may be present in one encounter and absent in the next.

Temporal knowledge graphs. MedTKG [10] constructs temporal KGs with time-stamped snapshots (event time only); Graphiti [11] implements bitemporal edges [24] but lacks clinical ontology alignment; TEO [13] and TCL [14] formalize Allen's relations [12] as annotation ontologies or modal operators rather than KG edge attributes. Uncertainty quantification in biomedical KGs [25] has not been connected to assertion taxonomies. Two structural gaps emerge: an *epistemic propagation gap* (assertion-detecting systems do not persist labels into KGs) and a *temporal integration gap* (temporal formalisms operate as annotation layers, not as KG edge attributes participating in retrieval). EpiKG closes both by carrying assertion and temporal metadata as first-class properties through every pipeline stage.

3 System Design

3.1 Overview

EpiKG transforms unstructured clinical narratives into an assertion-aware, temporally grounded knowledge graph exposed through graph-augmented retrieval. The pipeline comprises six stages: (1) document ingestion, (2) NLP extraction, (3) OMOP concept normalization [3], (4) clinical fact construction with epistemic metadata, (5) KG materialization with shared concept nodes, and (6) assertion-aware retrieval. The key invariant is that *epistemic assertion status* is a first-class attribute at every stage, not an NLP annotation discarded after extraction (Figure 1). Clinical concepts are stored once as shared global nodes; patient-specific semantics (assertion status, temporality, confidence) reside on edges, enabling cohort queries via edge scans.

3.2 Epistemic Assertion Schema

Clinical notes abound with qualified statements—"*no evidence of pneumonia*," "*possible CHF*," "*mother had breast cancer*"—yet standard representations discard these qualifiers. OMOP excludes negated conditions entirely [3]; FHIR limits assertion metadata to conditions. EpiKG defines a seven-value assertion taxonomy:

$$\alpha \in \{\text{Pres., Abs., Poss., Cond., Hypo., Fam.Hx., Hist.}\} \tag{1}$$

This extends the i2b2 six-class taxonomy [2] by separating HISTORICAL from FAMILY_HISTORY, a clinically important distinction that prior work conflates (definitions in Appendix Q). A rule-based classifier (101 scope-aware trigger patterns) assigns α with an associated confidence score; we chose rule-based classification for deterministic behavior and interpretability, though fine-tuned LLM classifiers [1] could be substituted. Intrinsic classifier evaluation is deferred; the end-to-end ClinicalBench ablation (C3→C4) serves as a functional test of assertion propagation. The label is carried through every stage: from `ExtractedMention` through ensemble merging, `ClinicalFact` records, KG edge properties, and retrieval scoring. To our knowledge, no prior system [1, 26] maintains this end-to-end invariant.

3.3 Tri-Temporal Knowledge Graph Model

Clinical events unfold across multiple time dimensions that prior systems model incompletely [10, 11]. Bi-temporal databases [24] distinguish valid time from transaction time; EpiKG adds a third dimension (NLP-asserted temporality) and stores all three on every KG edge:

Definition 1 (Tri-Temporal Edge). *An edge* $e = (s, p, o, \alpha, \boldsymbol{\tau}_v, \boldsymbol{\tau}_t, \tau_a, r, c)$ *where:*

- s, o *are source and target nodes;* p *is the predicate (one of 24 edge types);*
- $\alpha \in \mathcal{A}$ *is the assertion label (Eq. 1);*
- $\boldsymbol{\tau}_v = (\mathit{event\_date}, \mathit{valid\_from}, \mathit{valid\_to})$ *is the* ***valid time*** *interval—when the relationship held in the real world;*
- $\boldsymbol{\tau}_t = (\mathit{recorded\_at}, \mathit{doc\_date}, \mathit{created\_at})$ *is the* ***transaction time****—when it was recorded;*
- $\tau_a \in \{\text{CURRENT}, \text{PAST}, \text{FUTURE}\}$ *is the* ***NLP-asserted temporality****;*
- $r \in \mathcal{R}$ *is an Allen interval algebra relation [12];* $c \in [0, 1]$ *is temporal confidence.*

$\mathcal{R}$ comprises seven of Allen's 13 canonical relations [12] (merging symmetric pairs like Before/Meets) plus CONCURRENT and UNKNOWN (nine total; full mapping in Appendix R). Unlike TEO [13] and TCL [14], which use Allen's relations as annotation-ontology classes or modal operators, EpiKG stores r and c directly on materialized edges, enabling efficient temporal filtering during traversal.

3.4 Assertion-Aware Graph-Augmented Retrieval

Given a clinical question q and patient π, the retrieval pipeline: (1) extracts concepts from q via NLP + OMOP enrichment; (2) traverses 2–3 hops via bidirectional BFS over patient KG edges and OMOP vocabulary relationships (20M+ edges) via PostgreSQL CTEs (Appendix K), pruning edges with $c < 0.3$; (3) groups edges into four temporal views (event timeline, current state, historical, conflicts); (4) retrieves matching guideline sections; (5) scores edges by:

$$\text{score}(e) = c(e) + \mathbb{1}[\text{type}(e) \in Q_{\text{rel}}] \cdot 0.2 + \mathbb{1}[\tau_a(e) = \text{CURRENT}] \cdot 0.1 \tag{2}$$

where the bonuses for question-relevant edge types (0.2) and current temporality (0.1) were set by manual tuning on a development set of 20 questions. The surviving subgraph is serialized into structured text preserving assertion labels (e.g., "ABSENT: pneumonia"); and (6) composes graph evidence, temporal context, guidelines, and source documents into a single prompt.

3.5 Formal Epistemic Preservation

We formalize the epistemic invariant maintained by the EpiKG pipeline, building on the ontological principle that aboutness must be preserved across transformations [27]. The *epistemic state* of a clinical mention m is the tuple $e(m) = (c, \alpha, \xi, \tau)$: OMOP concept, 7-value assertion label, experiencer ($\{\text{PATIENT}, \text{FAMILY}\}$), and temporality ($\{\text{CURRENT}, \text{PAST}, \text{FUTURE}\}$); see Table 6 (Appendix) for notation summary. A pipeline *epistemically preserves* m if the output epistemic state equals the extraction state at every stage. We formalize (Appendix B) that an assertion-blind pipeline collapses the assertion distribution to a degenerate point mass, incurring entropy loss $\Delta H = H(A_c^\pi) \geq 0$ [28], and that assertion-faithful accuracy is bounded by $1 - f_{np}(c)$, where f_{np} is the fraction of non-present assertions. The empirical consequences are measured via category-stratified accuracy in Section 5.

Table 1: ClinicalBench conditions. C1/C4/C4g form the ablation ladder; C7 and C6 are bookend baselines.

ID	Condition	Retrieval	Assertion	Temporal
C7	Deterministic KG	KG lookup (no LLM)	Full	Tri-temporal
C1	LLM Alone	None	None	None
C4	+ Epistemic KG-RAG	Graph + Doc	Full (7-class)	Tri-temporal
C4g	+ Intent-Aware KG-RAG	Graph + Doc (type-specific)	Full (7-class)	Tri-temporal
C6	Long Context	All notes	None	None

3.6 Intent-Aware Retrieval (C4g)

The multi-hop BFS described above treats all questions uniformly, yet different clinical question types require fundamentally different graph operations. A *change* question requires cross-admission set differencing; a *current-state* question needs filtering to the most recent valid edges; a *historical* question must recover resolved conditions.

Intent classifier. A lightweight rule-based classifier maps each question to one of four intents—CHANGE, CURRENT_STATE, HISTORICAL, or DEFAULT—using question metadata with keyword-pattern fallback (Algorithm 1, Appendix S).

Change retrieval. For $\iota = \text{CHANGE}$, edges are partitioned by admission (`hadm_id`). The module computes set differences across admission pairs (k, k')—concepts *added* ($\mathcal{C}_{k'} \setminus \mathcal{C}_k$), *removed* ($\mathcal{C}_k \setminus \mathcal{C}_{k'}$), and *continued* ($\mathcal{C}_k \cap \mathcal{C}_{k'}$)—and returns structured evidence.

Current-state and historical retrieval. For CURRENT_STATE, the module filters to edges with $\tau_a = \text{CURRENT}$ or open-ended validity, deduplicates by concept, and emits "NOT FOUND" for missing concepts. For HISTORICAL, it selects edges with $\tau_a = \text{PAST}$ and augments with admission-based inference (concepts in earlier admissions but absent from the latest are labeled "resolved"). Each intent triggers a type-specific prompt template that structures evidence according to question semantics (Appendix U).

4 Benchmark Design

Existing clinical QA benchmarks evaluate factual recall (MedQA [29]) or agent task completion [30] but do not test handling of epistemic qualifiers or reasoning over longitudinal records of varying complexity. We introduce two complementary benchmarks built on MIMIC-IV [31].

4.1 ClinicalBench: Assertion-Sensitive Clinical QA

ClinicalBench comprises 400 gold-standard questions over 45 MIMIC-IV patients in two tasks: **Task A** (200 questions, negation-aware retrieval) and **Task B** (200 questions, temporal reasoning). Questions span 9 categories: *negation*, *conditional*, *uncertainty*, *family_history*, *sequence*, *current_state*, *duration*, *historical*, and *change*. Gold-standard answers were created through manual chart review and verified against source documents; a pilot physician adjudication ($n = 5$, Appendix N) assessed quality. Three ablation conditions progressively add system components (Table 1). Two bookend baselines—C6 (all notes concatenated) and C7 (deterministic KG lookup, no LLM)—bracket the design space.

Endpoints. The **pre-registered primary endpoint** is the hard longitudinal subset (*change* + *current_state* + *historical*, $n = 130$): C4g vs. C1 aggregate accuracy. The **secondary endpoint** is full 400-question ClinicalBench, C4g vs. C1. **Diagnostic endpoints** are per-category C1→C4g accuracy deltas and the C4→C4g transition on the hard subset.

What each transition isolates. C1→C4 isolates the *epistemic retrieval effect*: C4 adds graph traversal with assertion metadata, testing whether structured context helps or hurts. C4→C4g isolates the *intent-aware routing effect*: both carry full assertion metadata, but C4g dispatches question-type-specific graph traversals. C4g vs. C6 isolates the *structured-vs-unstructured retrieval effect* on the same questions.

4.2 SliceBench: Complexity-Stratified Longitudinal QA

SliceBench tests the hypothesis that KG-augmented retrieval provides increasing benefit as patient record complexity grows. We select 6 MIMIC-IV patients in three tiers: **Tier A** (2 patients,

Table 2: ClinicalBench primary results (400 questions, Claude Opus 4.6, keyword evaluator). C7 is model-independent (no LLM); remaining conditions use Opus 4.6. Best in **bold**.

	Condition	Accuracy	Δ vs C1
C7	Deterministic KG (no LLM)	3.8%	-45.7 pp
C1	LLM Alone	49.5%	—
C4	+ Epistemic KG-RAG	60.8%	$+11.3$ pp
C4g	+ Intent-Aware KG-RAG	**77.0%**	$+27.5$ pp
C6	Long Context (all notes)	41.8%	-7.7 pp

1–2 encounters), **Tier B** (2 patients, 5–10), and **Tier C** (2 patients, 15+). Each patient receives 24 questions (including hard longitudinal categories: cross-encounter medication timelines, problem list reconciliation, causal chain tracing), yielding 144 total. Five conditions (B0–B4) form a monotone context progression from parametric-only to full system (Appendix J). The critical comparison is B2→B3: both have the same documents, but B3 adds structured KG context with assertion metadata and temporal relations.

4.3 Evaluation Protocol

ClinicalBench uses a *deterministic keyword evaluator* that performs exact word-boundary matching of gold-standard assertion and temporal keywords against model answers (keyword lists in Appendix P). This evaluator is fully reproducible (no stochastic LLM judge) and is used for all ClinicalBench results. The primary answering model is Claude Opus 4.6; cross-model validation uses MedGemma 27B under the same evaluator.

SliceBench uses an *LLM-as-judge* protocol with separated answering and judging models to prevent self-evaluation bias [32]: Claude Sonnet 4.5 generates answers and Claude Opus 4.6 judges correctness.

Statistical reporting. We report BCa bootstrap 95% CIs ($n = 2{,}000$, seed 42) on SliceBench aggregate metrics (Table 5); CIs are computed at the question level. ClinicalBench results are reported as point estimates with within-run paired comparisons; CIs are omitted because the primary analysis targets condition deltas rather than absolute levels. A safety score with asymmetric weighting ($w = 2.0$ for false-positive assertion errors) captures clinical risk (Appendix D).

4.4 Physician Validation Protocol

To validate automated scoring, we preregister a two-physician blinded adjudication. Two board-certified emergency medicine physicians independently score a stratified sample of 168 questions under two conditions (C1 vs. C4g), blinded to condition assignment, using the same binary correctness rubric as the keyword evaluator. A third physician adjudicates disagreements; inter-rater reliability is reported as Cohen's κ. The protocol (Appendix O) was finalized before examining primary results. Results are pending.

5 Results

We evaluate EpiKG through two benchmarks: ClinicalBench tests assertion-sensitive and longitudinal reasoning, while SliceBench measures longitudinal QA across patient complexity tiers. ClinicalBench uses a deterministic keyword evaluator for reproducibility; SliceBench uses LLM-as-judge evaluation (Section 4.3). Claude Opus 4.6 is the primary answering model; MedGemma 27B provides cross-model validation. SliceBench reports BCa bootstrap 95% CIs ($n = 2{,}000$, seed 42) [33], resampled at the question level.

5.1 ClinicalBench: Primary Ablation

The bookend baselines bracket the design space: C7 (deterministic KG, no LLM) achieves 3.8%—only negation detection works (13.6%) via stored assertion metadata—confirming that an LLM is essential for reasoning over graph evidence. C6 (all notes concatenated) scores 41.8%, *worse* than C1 (-7.7 pp): brute-force long context is not a viable alternative, even with Opus's 200K token window. Uniform epistemic KG-RAG (C4) achieves 60.8% ($+11.3$ pp), showing that structured

Table 3: ClinicalBench per-category accuracy (%) across conditions (Claude Opus 4.6). C4g improves 8 of 9 categories over C1 with zero regressions. Best per row in **bold**.

Category	n	C6	C1	C4	C4g	C4g−C1
Negation	110	70.0	86.4	**94.5**	88.2	↑ 1.8 pp
Conditional	20	35.0	15.0	**50.0**	45.0	↑ 30.0 pp
Uncertainty	40	30.0	17.5	**52.5**	50.0	↑ 32.5 pp
Family hist.	30	43.3	3.3	50.0	**56.7**	↑ 53.3 pp
Sequence	40	0.0	**100.0**	97.5	**100.0**	= 0 pp
Current st.	50	46.0	40.0	32.0	**72.0**	↑ 32.0 pp
Duration	30	73.3	86.7	83.3	**93.3**	↑ 6.7 pp
Historical	50	0.0	6.0	12.0	**62.0**	↑ 56.0 pp
Change	30	43.3	10.0	23.3	**100.0**	↑ 90.0 pp
Overall	**400**	41.8	49.5	60.8	**77.0**	↑ 27.5 pp

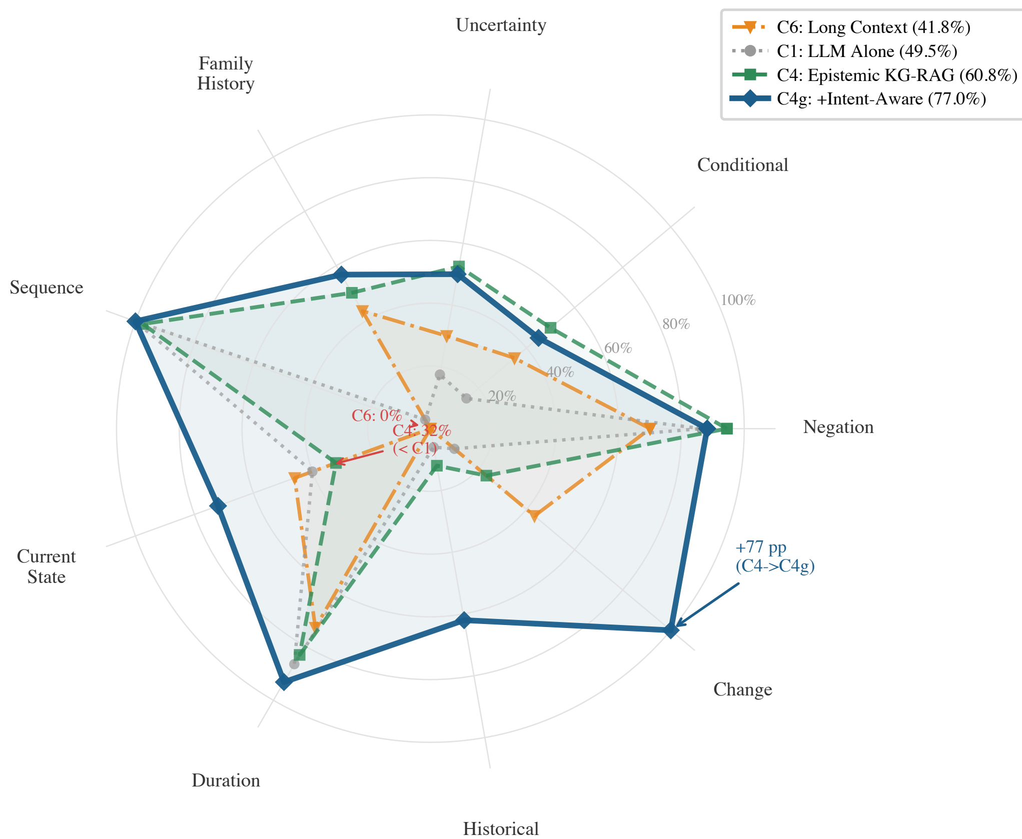

Figure 2: ClinicalBench per-category accuracy across four conditions (C6, C1, C4, C4g; Claude Opus 4.6). C6 (orange dash-dot) collapses on historical and sequence (both 0%); C4 (green dashed) improves negation and uncertainty but drops current state below C1; C4g (dark solid) improves 8 of 9 categories over C1 with no regressions.

retrieval with assertion metadata provides benefit even without intent-aware routing. Intent-aware KG-RAG (C4g) adds another +16.2 pp to reach 77.0%—the largest single-step improvement in the ablation, confirming that question-type-specific traversal is critical. The C4→C4g transition is most visible on change detection (23%→100%) and current state (32%→72%), where uniform retrieval injects irrelevant temporal context that intent-aware routing avoids.

Table 4: ClinicalBench cross-model results (400 questions, keyword evaluator). KG-RAG benefit is inversely related to parametric medical knowledge.

	Model	C1	C4g	Δ
Commercial	Claude Opus 4.6	49.5%	**77.0%**	+27.5 pp
Open	MedGemma 27B	52.5%	66.2%	+13.7 pp

Table 5: SliceBench overall results (144 questions, Claude Sonnet 4.5, judged by Opus 4.6). Sig: * denotes CI excluding zero.

	Condition	Score	95% CI	Δ vs B0
B0	LLM Alone	49.9%	[43.6%, 56.1%]	—
B1	Latest Note	73.4%	[67.8%, 78.7%]	+23.5 pp
B2	All Notes RAG	84.7%	[80.2%, 88.8%]	+34.8 pp *
B3	KG-RAG	86.9%	[82.8%, 90.8%]	+37.0 pp *
B4	Full System	87.6%	[83.7%, 91.2%]	+37.8 pp *

5.2 Category×Condition Interaction

Table 3 and Figure 2 reveal a strong *category×condition interaction*: intent-aware KG-RAG improves 8 of 9 categories with zero regressions—sequence is already at 100% under C1.

Where intent-aware routing excels. The largest gains occur on categories requiring cross-admission synthesis: change (+90 pp), historical (+56 pp), family history (+53 pp), uncertainty (+32.5 pp), and current state (+32 pp). C6 fails catastrophically on historical (0%) and sequence (0%) despite Opus's 200K window, confirming that long context cannot substitute for structured retrieval.

5.3 Cross-Model Validation

The KG-RAG benefit is model-dependent and inversely related to parametric medical knowledge (Table 4). Both models benefit from KG-RAG, but the magnitude differs: Opus gains +27.5 pp while MedGemma gains +13.7 pp.

This pattern is most visible on historical questions: MedGemma C4g achieves 46% on historical (vs. 0% for C1), while Opus C4g reaches 62% (vs. 6% for C1). The general-purpose model gains more from structured retrieval because it lacks the parametric medical knowledge that MedGemma acquired through fine-tuning. Intent-aware KG-RAG thus functions as a *medical knowledge compensator*: the structured epistemic context partially substitutes for parametric medical knowledge, providing the largest benefit to models that need it most. Per-category cross-model breakdowns appear in Appendix Table 8.

5.4 SliceBench

The incremental KG layer (B2→B3) contributes +2.2 pp overall (CI: [-1.5 pp, +5.9 pp]), not reaching aggregate significance, but tier-stratified estimates show a complexity-dependent trend: +0.6 pp (Tier A), +1.0 pp (Tier B), +5.0 pp (Tier C, 15+ encounters), consistent with KG context being most valuable when document volume overwhelms chunk-level retrieval (Appendix L).

5.5 Physician Adjudication

A two-physician blinded adjudication ($n = 168$ questions × 2 conditions, Cohen's κ for inter-rater reliability) is preregistered to validate automated scoring; the protocol (Appendix O) was designed before examining results. *Results are pending.*

6 Discussion and Limitations

Why long context fails. Brute-force long context (C6) scores −7.7 pp below LLM alone, even with Opus's 200K token window. C6 catastrophically fails on historical (0%) and sequence (0%): concatenating all notes chronologically buries temporal distinctions that are critical for these categories. Historical questions require distinguishing resolved from active conditions across admissions; sequence questions require strict temporal ordering—both are implicit in raw text but explicit in the epistemic KG. C6 handles change detection reasonably (43%) because the relevant information appears explicitly in the text, but KG-RAG achieves 100% (+57 pp). Uniform KG-RAG (C4) also underperforms intent-aware routing: 60.8% vs. C4g's 77.0%, with current state (32%) actually *below* C1 (40%), consistent with GraphRAG-Bench [34]: uniform graph traversal can inject context that overrides correct parametric reasoning.

Model-dependent KG benefit. The most striking finding is that KG-RAG benefit is inversely related to parametric medical knowledge (Table 4). Opus, a general-purpose model, gains $+28$ pp from intent-aware KG-RAG; MedGemma 27B, fine-tuned on medical corpora, gains $+14$ pp. Both models benefit, but the magnitude differs: Opus starts at 50% on C1 while MedGemma starts at 53%, yet Opus's absolute C4g performance (77%) exceeds MedGemma's (66%) because the epistemic KG context compensates more effectively for knowledge the general-purpose model lacks. Intent-aware KG-RAG thus functions as a *medical knowledge compensator*: the structured epistemic context partially substitutes for parametric medical knowledge, providing the largest benefit to models that need it most. This has practical implications: the same RAG system may help or hurt differently depending on the downstream model, and evaluation on a single model understates or overstates the system's value.

When KG context matters. The complexity-dependent SliceBench effect (Tier C: $+5.0$ pp vs. Tier A: $+0.6$ pp) suggests structured epistemic context is most valuable when: (a) document volume overwhelms chunk-level retrieval, (b) questions require cross-encounter synthesis where the same concept appears with different assertion states, and (c) assertion status evolves across encounters. C4g's intent-aware routing provides direct evidence: the "change" category jumps from 10% (C1) to 100% ($+90$ pp), confirming that *targeted graph traversal strategy*, not text access, is the key factor.

Uncertainty remains weakest. Uncertainty improves from 17.5% to 50.0% ($+32.5$ pp), but absolute accuracy remains low. Possible and suspected conditions are linguistically subtle ("consider statin," "may have CHF") and require pragmatic inference beyond keyword matching.

Limitations. Sample size is modest: ClinicalBench covers 45 patients (400 questions) and SliceBench 6 patients (144 questions), both from MIMIC-IV [31]. The B2→B3 CI crosses zero ([-1.5 pp, +5.9 pp]); tier-level estimates ($n = 2$ patients per tier) are hypothesis-generating only. Model-dependent effects mean headline gains ($+28$ pp) are specific to Opus + C4g; MedGemma shows $+14$ pp with smaller absolute improvement. The keyword evaluator may miss semantically correct answers; physician adjudication (Section 5.5) will address this. This is an ablation study; comparisons with medical RAG systems [5, 6, 15] were not performed because these target factoid QA rather than patient-level longitudinal reasoning. Bootstrap CIs are question-level; patient-level clustering is planned. MedGemma run-to-run variance (± 10 pp) arises from GPU non-determinism in 4-bit GGUF inference. Over-reliance on KG-driven summaries risks anchoring bias if NLP extraction introduces systematic errors; single-site evaluation (US academic center) leaves generalization unvalidated (Appendix V).

Conclusion. The epistemic propagation gap is a real and measurable phenomenon: brute-force long context degrades accuracy by 8 pp, while intent-aware epistemic KG-RAG improves it by 28 pp. EpiKG addresses this gap through end-to-end epistemic preservation combined with intent-aware retrieval that matches graph traversal strategy to question semantics. The benefit is category-specific: change detection rises from 10% to 100% ($+90$ pp), historical from 6% to 62% ($+56$ pp), family history from 3% to 57% ($+53$ pp)—8 of 9 categories improve with zero regressions. The benefit is also model-dependent: KG-RAG functions as a medical knowledge compensator, providing the largest gains to models that need it most. Neither more context nor structured data alone suffices—C6 and C7 (3.8%) bracket the design space—confirming that the category$\times$condition interaction, not the aggregate, is the right lens for clinical KG-RAG evaluation.

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A Notation Summary

Symbol	Meaning
$\alpha \in \mathcal{A}$	Assertion label (7 values, Eq. 1)
$\boldsymbol{\tau}_v$	Valid time (event date, valid from/to)
$\boldsymbol{\tau}_t$	Transaction time (recorded at, doc date, created at)
τ_a	NLP-asserted temporality (Current / Past / Future)
$r \in \mathcal{R}$	Temporal relation (9 values, Table 16)
c	Confidence score $\in [0, 1]$
ξ	Experiencer (Patient / Family)
$e(m)$	Epistemic state tuple (c, α, ξ, τ)
q, π	Clinical question, patient
ι	Question intent (Change / Current_State / Historical / Default)
$f_{np}(c)$	Fraction of non-present assertions for concept c

Table 6: Key notation used throughout the paper.

B Formal Epistemic Preservation

We formalize the epistemic invariant maintained by the EpiKG pipeline, building on the principle that aboutness—the relationship between information artifacts and the entities they represent—must be preserved across transformations [27].

Definition 2 (Epistemic State)**.** *For a clinical mention m, the* epistemic state *is the tuple $e(m) = (c, \alpha, \xi, \tau)$, where c is the OMOP concept identifier, $\alpha \in \mathcal{A}$ is the 7-value assertion label (Eq. 1), $\xi \in \{\text{PATIENT}, \text{FAMILY}\}$ is the experiencer, and $\tau \in \{\text{CURRENT}, \text{PAST}, \text{FUTURE}\}$ is the temporality. A pipeline P* epistemically preserves *mention m if $P(e(m)) = e(m)$; that is, the epistemic state at the output of every pipeline stage is identical to the state at extraction.*

Proposition 3 (Assertion Entropy Loss)**.** *For concept c in patient π's record, let A_c^π denote the assertion distribution with empirical frequencies $p(\alpha_i) = n_i / \sum_j n_j$ over the $|\mathcal{A}|$ assertion classes. The assertion entropy is*

$$H(A_c^\pi) = -\sum_i p(\alpha_i) \log p(\alpha_i) \ . \tag{3}$$

An assertion-blind pipeline collapses all mentions to $\alpha = \text{PRESENT}$, yielding a degenerate distribution with $H = 0$. The information loss is $\Delta H = H(A_c^\pi) \geq 0$ [28], with $\Delta H > 0$ strictly whenever any mention carries a non-present assertion.

Proof. Under the assertion-blind mapping $\phi : \alpha_i \mapsto \text{PRESENT}$ for all i, the output distribution assigns probability 1 to PRESENT and 0 to all other classes, so $H(\phi(A_c^\pi)) = 0$. By non-negativity of Shannon entropy, $\Delta H = H(A_c^\pi) - 0 = H(A_c^\pi) \geq 0$, with equality iff all mentions already carry $\alpha = \text{PRESENT}$. □

Corollary 4 (Faithfulness Bound)**.** *Let $f_{np}(c)$ denote the fraction of mentions of concept c carrying a non-present assertion ($\alpha \neq \text{PRESENT}$). Without assertion labels, the maximum assertion-faithful accuracy for any downstream task conditioned on c is bounded by $1 - f_{np}(c)$. In clinical records where negated and uncertain mentions are prevalent—e.g., "no pneumonia," "possible CHF"—this bound can be substantially below 1.*

Proof. Without assertion labels, any predictor must assign a single assertion class to all mentions of c. Choosing PRESENT (the majority class in clinical text) yields accuracy $1 - f_{np}(c)$; the $f_{np}(c)$ non-present mentions are necessarily misclassified. □

C Gap Analysis

Partial marks indicate limited coverage: GFM-RAG, MedRAG, and KARE build population-level or hierarchical KGs (not per-patient longitudinal graphs); KARE uses KG-augmented retrieval but

Table 7: Capability comparison across representative systems. ✓ = supported, ○ = partial, × = not supported.

System	Patient KG	OMOP mapping	Assertion (7-class)	Tri-temporal	Assert.-aware RAG	Experiencer	Allen's algebra	Multi-hop (≥3)
DoctorRAG [15]	×	×	×	×	×	×	×	×
GFM-RAG [5]	○	×	×	×	×	×	×	✓
MedRAG [16]	○	×	×	×	×	×	×	○
KARE [6]	○	×	×	×	○	×	×	✓
Multi-LLM KG [18]	×	✓	○	×	×	×	×	×
MedTKG [10]	✓	×	×	○	×	×	×	✓
Graphiti [11]	×	×	×	○	×	×	×	✓
EpiKG (ours)	✓	✓	✓	✓	✓	✓	✓	○

without assertion awareness; Multi-LLM KG models uncertainty via entropy but not the full assertion taxonomy; MedTKG and Graphiti use temporal annotations but not tri-temporal models. EpiKG's partial mark on multi-hop traversal ($\geq$ 3 hops) reflects a deliberate architectural trade-off: PostgreSQL CTE-based traversal provides ACID compliance but degrades beyond 2 hops (Appendix K).

D Safety Score

The safety score $S = 1 - \frac{1}{N}\sum_{i=1}^{N} w_i \cdot \mathbb{1}[\hat{y}_i \neq y_i]$ applies $w_i = 2.0$ for false-positive assertion errors (reporting a negated or absent condition as present) and $w_i = 1.0$ otherwise, capturing asymmetric clinical risk where acting on a falsely affirmed condition is more dangerous than missing a present one. We treat $w = 2.0$ as a reasonable default; sensitivity to this choice is a direction for future work.

E Cross-Model Per-Category Results

Table 8: ClinicalBench per-category accuracy (%) by model under C1 and C4g (intent-aware KG-RAG). Opus improves 8/9 categories; MedGemma improves 5/9 but regresses on negation, conditional, and duration.

		Claude Opus 4.6			MedGemma 27B		
Category	n	C1	C4g	Δ	C1	C4g	Δ
Negation	110	86.4	88.2	↑1.8 pp	93.6	76.4	↓17.2 pp
Conditional	20	15.0	45.0	↑30.0 pp	45.0	35.0	↓10.0 pp
Uncertainty	40	17.5	50.0	↑32.5 pp	10.0	32.5	↑22.5 pp
Family hist.	30	3.3	56.7	↑53.3 pp	3.3	43.3	↑40.0 pp
Sequence	40	100.0	100.0	=	100.0	100.0	=
Current st.	50	40.0	72.0	↑32.0 pp	36.0	58.0	↑22.0 pp
Duration	30	86.7	93.3	↑6.7 pp	96.7	90.0	↓6.7 pp
Historical	50	6.0	62.0	↑56.0 pp	0.0	46.0	↑46.0 pp
Change	30	10.0	100.0	↑90.0 pp	20.0	96.6	↑76.7 pp
Overall	**400**	**49.5**	**77.0**	**↑27.5** pp	**52.5**	**66.2**	**↑13.7** pp

F MedGemma Full Per-Category Results

Category	n	**C7**	**C6**	**C1**	**C4g**
Negation	110	13.6	72.7	93.6	76.4
Conditional	20	0.0	70.0	45.0	35.0
Uncertainty	40	0.0	32.5	10.0	32.5
Family History	30	0.0	63.3	3.3	43.3
Sequence	40	0.0	87.5	100.0	100.0
Current State	50	0.0	16.0	36.0	58.0
Duration	30	0.0	100.0	96.7	90.0
Historical	50	0.0	18.0	0.0	46.0
Change	30	0.0	73.3	20.0	96.6
Overall	**400**	**3.8**	**57.5**	**52.5**	**66.2**

Table 9: MedGemma 27B ClinicalBench per-category accuracy (%, deterministic keyword evaluator) for four conditions. MedGemma gains $+14$ pp overall under C4g, with strongest improvements on change ($+77$ pp) and historical ($+46$ pp).

G Experiencer Attribute Propagation Ablation

The `experiencer` attribute distinguishes patient conditions from family member conditions; without it, the graph conflates the two, causing family history misattribution.

Table 10: Impact of experiencer attribute propagation on Opus C4 (400 questions, prior evaluator run). Categories with no change omitted.

Category	Pre-fix	Post-fix	Δ
Family history	63.3%	**73.3%**	$+10.0$ pp
Uncertainty	50.0%	**55.0%**	$+5.0$ pp
Historical	34.0%	38.0%	$+4.0$ pp
Current state	46.0%	48.0%	$+2.0$ pp
Overall C4	**68.2%**	**70.2%**	$+2.0$ pp

The fix improved family history by $+10.0$ pp with zero regressions on guard categories (negation, conditional, duration, sequence all unchanged), confirming that the experiencer attribute is load-bearing for assertion-sensitive reasoning.

H SliceBench Figures

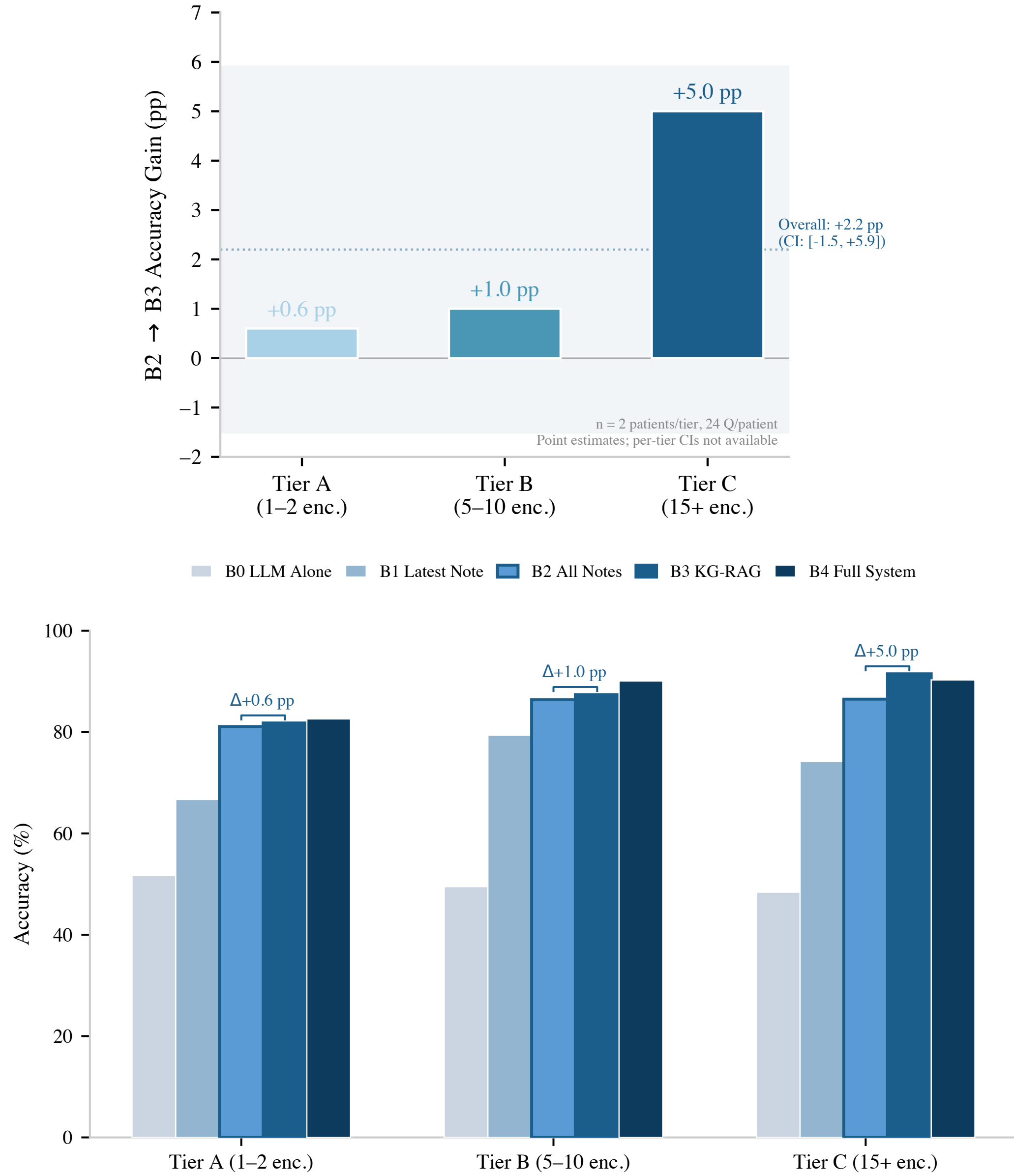

Figure 3: SliceBench results ($n = 2$ patients per tier, 24 questions each; point estimates without per-tier CIs). **Top:** The KG layer's incremental benefit (B2→B3) shows a directional, complexity-dependent trend: $+0.6$ pp (Tier A) to $+5.0$ pp (Tier C). **Bottom:** Full condition progression by tier. The overall B2→B3 delta is $+2.2$ pp (CI: [-1.5 pp, +5.9 pp]).

I Per-Category Delta Chart

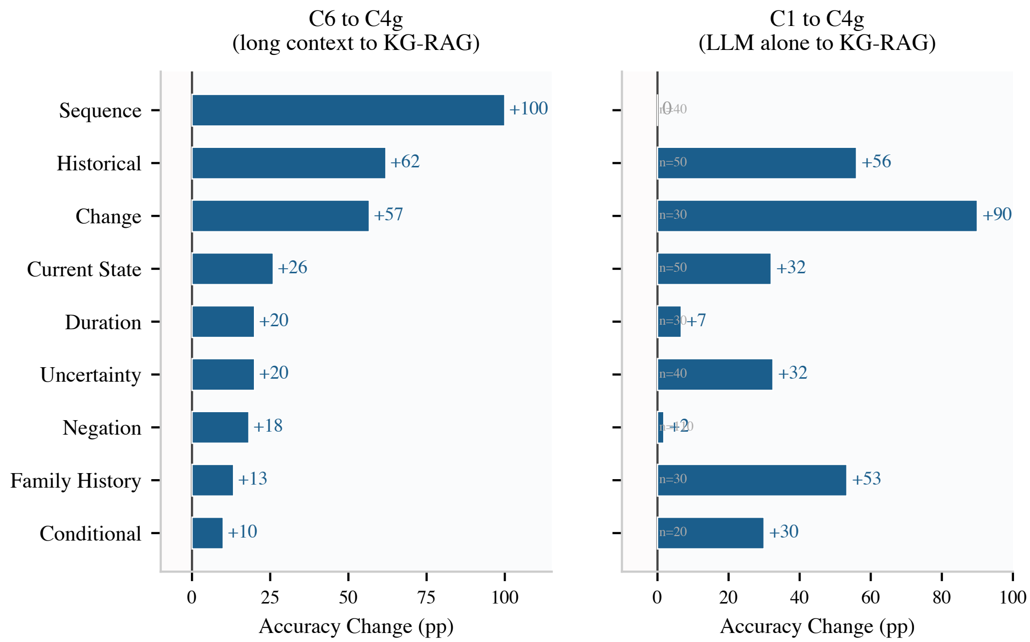

Figure 4: Per-category accuracy change (Claude Opus 4.6) for two comparisons. **Left:** C6→C4g (structured vs. unstructured retrieval): KG-RAG outperforms long context on all 9 categories, with the largest gains on sequence (+100 pp) and historical (+62 pp). **Right:** C1→C4g (LLM alone to KG-RAG): gains on 8 of 9 categories—change (+90 pp), historical (+56 pp), family history (+53 pp)—with zero regressions.

J SliceBench Conditions

Table 11: SliceBench conditions. B0–B4 form a monotone context progression.

ID	Condition	Context Provided to LLM
B0	LLM Alone	Question only (parametric knowledge)
B1	Latest Note	Most recent clinical note
B2	All Notes RAG	All notes, retrieved by relevance
B3	KG-RAG	Knowledge graph context + all notes
B4	Full System	KG-RAG + guidelines + calculators

K System Implementation Details

LLM baseline. On MedQA-USMLE (965 questions), Claude Opus 4.5 achieves 81.6% accuracy (5.1 pp below GPT-4 at 86.7% and Med-PaLM 2 at 86.5% [30]), establishing that the Claude model family—Sonnet 4.5 (SliceBench answerer) and Opus 4.6 (ClinicalBench primary, SliceBench judge)—provides a reasonable LLM baseline (Table 12).

Model	**Accuracy**	**Step 1**	**N**
EpiKG (LLM alone)	81.6%	79.9%	965
GPT-4 (2023)	86.7%	—	1,273
Med-PaLM 2 (2023)	86.5%	—	1,273
Claude 3 Opus (2024)	78.0%	—	1,273

Table 12: MedQA-USMLE results. Our LLM-alone baseline is competitive.

KG scale and traversal latency. The deployed system processes 145 documents across 85 patients, materializing 3,100 KG nodes and 8,803 edges. Graph traversal latency is sub-millisecond at this scale: 0.57 ms (1-hop), 0.75 ms (2-hop). The full system integrates 201 clinical calculators, 1,202 guideline sections, 20M+ OMOP vocabulary relationships, 101 assertion trigger patterns, and 24 edge types across 13 node types.

Multi-hop traversal. On the DR.KNOWS diagnostic reasoning benchmark [35], which measures KG traversal accuracy using PostgreSQL-backed clinical data, EpiKG achieves 0.420 overall (50% at 1-hop, 25% at 2-hop, 0% at 3-hop). Multi-hop degradation reflects a deliberate architectural trade-off: PostgreSQL CTE-based traversal provides ACID compliance but scales poorly beyond 2 hops.

L SliceBench Tier-Stratified Results

Table 13: SliceBench results stratified by patient complexity tier. B2→B3 Δ shows the incremental KG contribution.

		Score (%)					
Tier	Encounters	B0	B1	B2	B3	B4	B2→B3 Δ
A	1–2 notes	51.7	66.7	81.1	81.8	82.6	+0.6 pp
B	5–10 notes	49.5	79.4	86.4	87.4	90.1	+1.0 pp
C	15+ notes	48.4	74.2	86.5	91.5	90.3	+5.0 pp

M ClinicalBench Full Per-Category Results (Opus)

Category	n	**C7**	**C6**	**C1**	**C4**	**C4g**
Negation	110	13.6	70.0	86.4	94.5	88.2
Conditional	20	0.0	35.0	15.0	50.0	45.0
Uncertainty	40	0.0	30.0	17.5	52.5	50.0
Family History	30	0.0	43.3	3.3	50.0	56.7
Sequence	40	0.0	0.0	100.0	97.5	100.0
Current State	50	0.0	46.0	40.0	32.0	72.0
Duration	30	0.0	73.3	86.7	83.3	93.3
Historical	50	0.0	0.0	6.0	12.0	62.0
Change	30	0.0	43.3	10.0	23.3	100.0
Overall	**400**	**3.8**	**41.8**	**49.5**	**60.8**	**77.0**

Table 14: Complete ClinicalBench per-category accuracy (%, Claude Opus 4.6, deterministic keyword evaluator) for all conditions. C7 is model-independent. C4→C4g: change jumps from 23% to 100% and current state from 32% to 72%, isolating intent-aware routing.

N Human Validation Pilot

To assess LLM-as-judge reliability, we conducted a pilot validation study with a single board-certified physician who scored system outputs on a 4-point scale without knowledge of condition labels. The pilot ($n = 5$) revealed overly strict automated scoring on uncertainty and sequence questions. Gold standards were fully correct in 3/5 cases; model answers were partially correct or better in 3/5 cases where the auto-scorer marked them incorrect. A full two-physician blinded adjudication is preregistered (Section 5.5, Appendix O).

O Physician Validation Protocol

Design. Two board-certified emergency medicine physicians independently score ClinicalBench answers under two conditions (C1 and C4g) using the same binary correctness rubric as the deterministic keyword evaluator. Reviewers are blinded to condition assignment. The sample comprises 168 questions stratified across all 9 categories, with both conditions scored per question (336 total scored pairs).

Inter-rater agreement. Cohen's κ is computed between the two reviewers on the full 336 scored pairs. A third board-certified physician adjudicates all disagreements.

Endpoints. (1) Human–keyword evaluator concordance rate (fraction of automated scores confirmed by physicians). (2) Human-adjudicated C4g vs. C1 accuracy delta. (3) Category-level human accuracy.

Disclosure. Both primary reviewers are co-authors. We report κ to demonstrate independent judgment and disclose the relationship in the main text.

Status. This protocol was designed before examining primary results. *Results are pending.*

P Reproducibility Details

Models. ClinicalBench primary ablation: Claude Opus 4.6 (claude-opus-4-20250514, keyword evaluator). Cross-model: MedGemma 27B (gemma3:27b, 4-bit GGUF) via Ollama; same evaluator. SliceBench: Claude Sonnet 4.5 (claude-sonnet-4-5-20250929) answers, Opus 4.6 judges. Temperature 0 throughout; 4-bit quantization introduces GPU non-determinism (± 10 pp run-to-run for MedGemma), controlled via within-run paired comparisons.

Evaluation. The deterministic keyword evaluator performs exact word-boundary matching (regex `\bKEYWORD\b`) against gold-standard assertion and temporal keywords. Per-category keyword sets: negation ("no," "denies," "absent," "negative"), uncertainty ("possible," "suspected," "may," "likely," "consider"), temporal ("before," "after," "during," "changed," "new"), plus category-specific terms. Evidence preambles (echoed graph context) are stripped before matching.

Retrieval. C2's vanilla RAG uses TF-IDF over chunked clinical notes (512-token chunks, 64-token overlap), retrieving the top-5 chunks by cosine similarity. ClinicalBench conditions map to SliceBench: C1$\approx$B0, C2$\approx$B2, C4$\approx$B3, C5$\approx$B4.

Bootstrap. $n = 2{,}000$ resamples, seed 42, BCa method [33].

Data. De-identified MIMIC-IV clinical notes accessed under a PhysioNet Credentialed Health Data Use Agreement (v1.5.0).

Compute. ClinicalBench Opus C1/C4g: $\sim$22min each (API); Opus C6: $\sim$3.5h (API); C7: $<$ 1min (no LLM); MedGemma C1–C4g: $\sim$3.6h (single GPU); MedGemma C6: $\sim$1.5h; SliceBench: $\sim$2h (API).

Evaluator evolution. Key refinements: (i) substring to word-boundary matching, (ii) evidence preamble stripping. All main-text numbers use the final version.

Reproducibility package. A standalone reproducibility package is included as supplementary material (`epikg-benchmark/`). It contains all 400 ClinicalBench questions with gold answers, raw model predictions for all six condition$\times$model combinations (Opus C1/C4/C4g/C6, MedGemma C1/C4g), and the 284-line deterministic keyword evaluator. Running the evaluator requires only Python 3.10+ with no external dependencies. All six accuracy numbers in Tables 2 and 4 are independently reproducible from this package. MIMIC-IV patient identifiers are included so that PhysioNet-credentialed reviewers can trace predictions back to source clinical notes.

Q Assertion Category Definitions

Assertion	Definition	Example
PRESENT	Condition currently affirmed	"has diabetes"
ABSENT	Condition explicitly negated	"denies chest pain"
POSSIBLE	Condition suspected, not confirmed	"possible pneumonia"
CONDITIONAL	Contingent on specific circumstances	"if febrile, start antibiotics"
HYPOTHETICAL	Discussed as a scenario	"would need dialysis if…"
FAMILY_HISTORY	Attributed to a family member	"mother had breast cancer"
HISTORICAL	Previously true, not necessarily current	"former smoker"

Table 15: The 7-value assertion taxonomy, extending i2b2 by separating HISTORICAL from FAMILY_HISTORY.

R Temporal Relation Mapping

The nine temporal relations $\mathcal{R}$ used on KG edges are derived from Allen's 13 canonical interval relations [12] by merging symmetric pairs.

$\mathcal{R}$ **value**	**Allen source(s)**	**Meaning**
BEFORE	Before, Meets	A ends before B starts
AFTER	After, Met-by	A starts after B ends
DURING	During	A entirely within B
CONTAINS	Contains	A entirely contains B
OVERLAPS	Overlaps, Overlapped-by	A and B partially overlap
STARTS	Starts, Started-by	A and B share start time
FINISHES	Finishes, Finished-by	A and B share end time
CONCURRENT	Equals	A and B have same interval
UNKNOWN	—	Relation undetermined

Table 16: Mapping from Allen's 13 interval relations to the 9 temporal relation values stored on KG edges.

S Intent-Aware Routing Algorithm

The C4g classifier is rule-based, mapping questions to one of four intents using keyword patterns (e.g., "changed", "new since" trigger CHANGE; "currently", "active problem" trigger CURRENT_STATE). Algorithm 1 details the full retrieval procedure.

Algorithm 1: Intent-Aware Retrieval (C4g)

Input: Question q, patient π
Output: Structured evidence E

$\mathcal{C} \leftarrow \textsc{ExtractConcepts}(q)$; `// NLP + OMOP enrichment`
$\iota \leftarrow \textsc{ClassifyIntent}(q)$; `//` $\in \{\textsc{Change}, \textsc{CurrSt}, \textsc{Hist}, \textsc{Default}\}$
if $\iota = \textsc{Change}$ **then**
  Partition edges by admission: $\mathcal{E}_k \leftarrow \{e \mid \texttt{hadm\_id}(e) = k\}$;
  foreach *admission pair* (k, k') *with* $k < k'$ **do**
    $\mathcal{A} \leftarrow \mathcal{C}_{k'} \setminus \mathcal{C}_k$; $\mathcal{R} \leftarrow \mathcal{C}_k \setminus \mathcal{C}_{k'}$; $\mathcal{S} \leftarrow \mathcal{C}_k \cap \mathcal{C}_{k'}$;
  $E_g \leftarrow \textsc{FormatChange}(\mathcal{A}, \mathcal{R}, \mathcal{S})$;
else if $\iota = \textsc{CurrSt}$ **then**
  $E_g \leftarrow \textsc{FilterEdges}(\pi, \mathcal{C},\ \tau_a = \textsc{Current} \vee \text{open validity})$;
  Deduplicate by concept; emit "NOT FOUND" for missing $c \in \mathcal{C}$;
else if $\iota = \textsc{Hist}$ **then**
  $E_g \leftarrow \textsc{FilterEdges}(\pi, \mathcal{C},\ \tau_a = \textsc{Past})$;
  Augment: concepts in earlier admissions but absent from latest $\rightarrow$ "resolved";
else
  $E_g \leftarrow \textsc{BidirectionalBFS}(\pi, \mathcal{C},\ \text{hops} = 2\text{–}3,\ c_{\min} = 0.3)$;
$E_d \leftarrow \textsc{RetrieveDocuments}(\pi, \mathcal{C})$;
return $\textsc{Compose}(E_g, E_d,\ \text{template}(\iota))$;

T Bookend and Diagnostic Conditions

C5 (Full System). C5 extends C4 with guideline retrieval (1,202 sections) and clinical calculators (201, e.g., CHA_2DS_2-VASc, MELD). In a prior evaluator run, C5 achieved 68.2% overall, below the prior-run C1 and C4, likely because non-relevant components dilute the context window.

C6 (Long Context). All patient documents are concatenated chronologically and presented to the LLM. For Opus (200K token window), all notes fit; for MedGemma (8K window), later documents are truncated. C6 achieves 41.8% (Opus) and 57.5% (MedGemma) overall. Both models catastrophically fail on historical (0% Opus, 18% MedGemma), confirming that brute-force long context cannot substitute for structured retrieval regardless of context window size.

C7 (Deterministic KG). KG edges matching query concepts are returned directly without LLM reasoning. C7 achieves 3.8% overall (only negation at 13.6% via stored assertion metadata), confirming that structured data without an LLM reasoner cannot answer clinical questions.

U Illustrative Retrieval Example

To illustrate why intent-aware routing matters, consider a change question: *"What medications changed between this patient's first and second admissions?"*

C1 (LLM alone). The model sees only the current note, which mentions metoprolol, atorvastatin, and "discontinued lisinopril." Without prior admission records, it cannot determine what was *added* vs. *continued*, and may hallucinate prior medication lists.

C4g (intent-aware KG-RAG). The intent classifier routes to CHANGE retrieval, which partitions KG edges by `hadm_id` and computes set differences:

- **Added** (admission 2 only): atorvastatin 40mg [PRESENT]
- **Removed** (admission 1 only): lisinopril 10mg [PRESENT→ABSENT]
- **Continued**: metoprolol 25mg [PRESENT, both admissions]

The assertion labels (PRESENT, ABSENT) disambiguate: "discontinued lisinopril" is not a current medication but a historical one whose status changed—exactly the distinction the epistemic schema preserves.

V Ethics Statement

This work uses de-identified clinical data from MIMIC-IV [31] under a PhysioNet Credentialed Health Data Use Agreement. No patient re-identification was attempted. The system is designed for clinical decision *support*, not autonomous clinical decision-making.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: **Yes.**

   Justification: The abstract and introduction state four specific contributions (Section 1), each supported by empirical evidence in Sections 5.1–5.4. SliceBench claims include bootstrap CIs; ClinicalBench claims use within-run paired comparisons (Section 4.3). Claims are scoped with "to our knowledge" qualifiers where appropriate.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: **Yes.**

   Justification: Section 6 explicitly discusses limitations including small sample sizes (6 patients for SliceBench, 2 per tier), single-site data (MIMIC-IV), the B2→B3 confidence interval crossing zero, model-dependent effects, and pending physician adjudication.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (correct) proof?

   Answer: **Yes.**

   Justification: Proposition 3 (assertion entropy loss) and Corollary 4 (faithfulness bound) in Appendix B state all assumptions explicitly. These are information-theoretic characterizations with all steps included.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper?

   Answer: **Yes.**

   Justification: Appendix P reports exact model versions (MedGemma 27B, Claude Sonnet 4.5, Claude Opus 4.6), temperature settings (0), bootstrap parameters (n = 2,000, seed 42, BCa method), and data source (MIMIC-IV v1.5.0 under PhysioNet DUA). Section 4 describes question generation and evaluation protocols.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results?

   Answer: **Yes.**

   Justification: A standalone reproducibility package is included as supplementary material (`epikg-benchmark/`) containing all 400 questions, raw predictions for six condition×model combinations, and the deterministic keyword evaluator. All main-text accuracy numbers are independently reproducible. MIMIC-IV source notes require separate PhysioNet credentialed access. Full system code will be released upon acceptance.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details necessary to understand the results?

   Answer: **Yes.**

   Justification: Section 4 specifies benchmark design, question counts, condition definitions, patient selection criteria, and evaluation methodology. Appendix P provides model versions and computational requirements.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly?

   Answer: **Yes.**

   Justification: Table 5 reports BCa bootstrap 95% CIs for all conditions. Per-tier results (Figure 3) are clearly labeled as point estimates without per-tier CIs due to small tier-level sample sizes ($n = 2$ patients per tier). The overall B2→B3 CI crossing zero is explicitly noted.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources needed to reproduce the experiments?

   Answer: **Yes.**

   Justification: Appendix P reports computational requirements: ClinicalBench Opus ≈22 min per condition (API), MedGemma ≈3.6 hours (single GPU), SliceBench ≈2 hours (API).

9. **Code of ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics?

Answer: **Yes.**

Justification: The study uses de-identified clinical data under an approved data use agreement (Appendix V). No patient re-identification was attempted. The system is designed for clinical decision *support*, not autonomous decision-making.

10. **Broader impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work?

    Answer: **Yes.**

    Justification: Section 6 discusses the potential for improved clinical decision support (positive) and risks of over-reliance on automated systems, context dilution, and the need for physician oversight (negative). Appendix V addresses data ethics.

11. **Safeguards**

    Question: Does the paper describe safeguards that have been put in place for responsible release of data or models?

    Answer: **Yes.**

    Justification: The system operates on de-identified data under PhysioNet DUA. It is designed as a decision support tool requiring physician oversight. Code release will include usage guidelines emphasizing the system is not validated for autonomous clinical use.

12. **Licenses for existing assets**

    Question: Are the creators or original owners of assets used in the paper properly credited and are the license and terms of use explicitly mentioned and properly respected?

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    Justification: MIMIC-IV is cited [31] and used under PhysioNet Credentialed Health Data Use Agreement. All benchmark frameworks and models are cited. OMOP CDM is an open standard.

13. **New assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: **Yes.**

    Justification: ClinicalBench and SliceBench are new benchmarks fully documented in Section 4, with question generation procedures, category definitions (Appendix Q), and evaluation protocols. Templates and scoring scripts will accompany the code release.

14. **Crowdsourcing and research with human subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots?

    Answer: **N/A.**

    Justification: No crowdsourcing was used. Physician adjudication (Appendix N) involves expert review by co-authors, not human subjects research.

15. **IRB approvals or equivalent**

Question: Does the paper describe potential risks and does it include the IRB approval number if applicable?

Answer: **N/A.**

Justification: MIMIC-IV is a de-identified public dataset; use under PhysioNet DUA does not require separate IRB approval per PhysioNet policy.

16. **Declaration of LLM usage**

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods?

Answer: **Yes.**

Justification: LLMs are core experimental components, not authoring aids. Claude Opus 4.6 is the primary answering model for ClinicalBench (Section 5.1); Claude Sonnet 4.5 answers SliceBench questions while Opus 4.6 judges (Section 5.4); MedGemma 27B provides cross-model validation (Section 5.3). Model versions, temperatures, and inference details are in Appendix P.